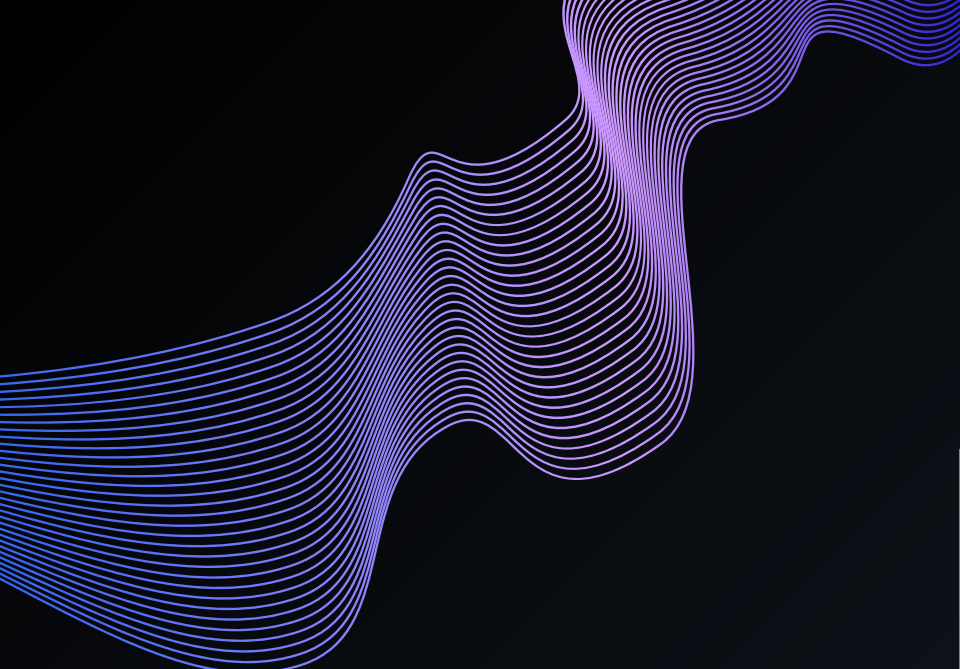
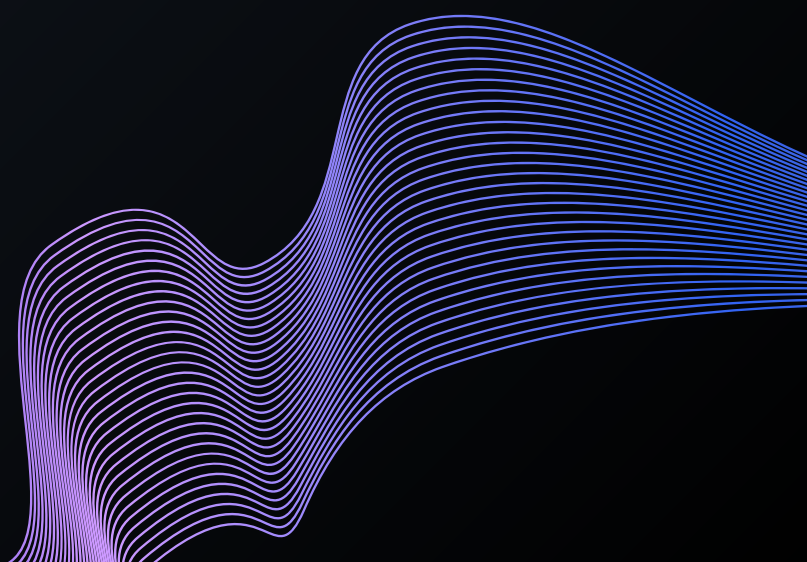
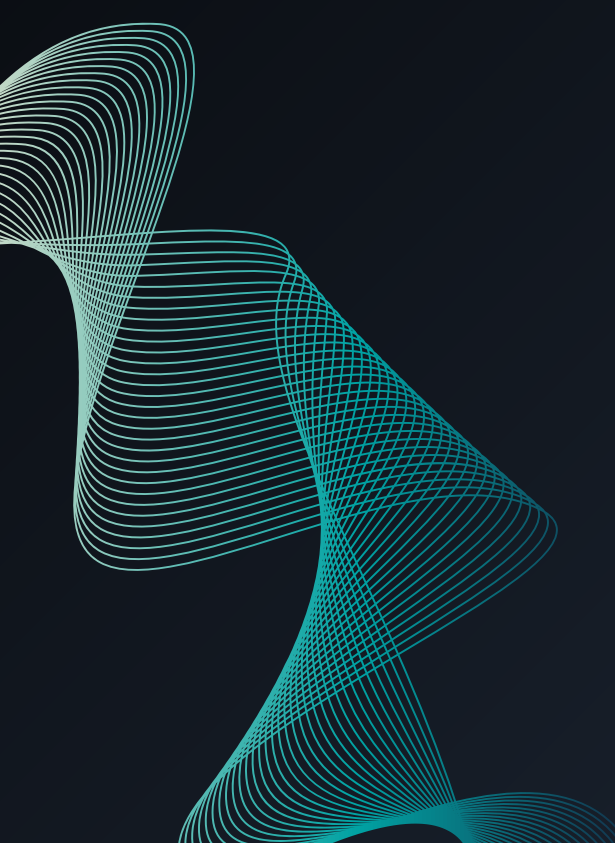




LINEAGE-TRACE

Reconstructing the Lineage of Deception

Made By: Ananya Palnitkar
Email: shivania.palnitkar@gmail.com



THE PROBLEM

- 11:42 PM A candidate-related claim appears in Hindi on a regional Telegram channel. By 11:58, it reaches X, reworded with a changed number and stripped context. By 12:15 AM, screenshots circulate on WhatsApp in three languages, each version more inflammatory than the last.
- Which version is the original? Which one is driving the spread? Lineage Trace's answer: cluster → diff → danger-score → topology. The WhatsApp version flags highest-risk, spread pattern reads as coordinated.

Existing tools tell you whether a claim is true or whether a narrative is trending, but not how the claim changed as it travelled.

For newsrooms, political monitoring teams, and reputation teams, this creates a critical blind spot. Manual tracing is reactive and too slow, often identifying the origin only after the narrative has already saturated.

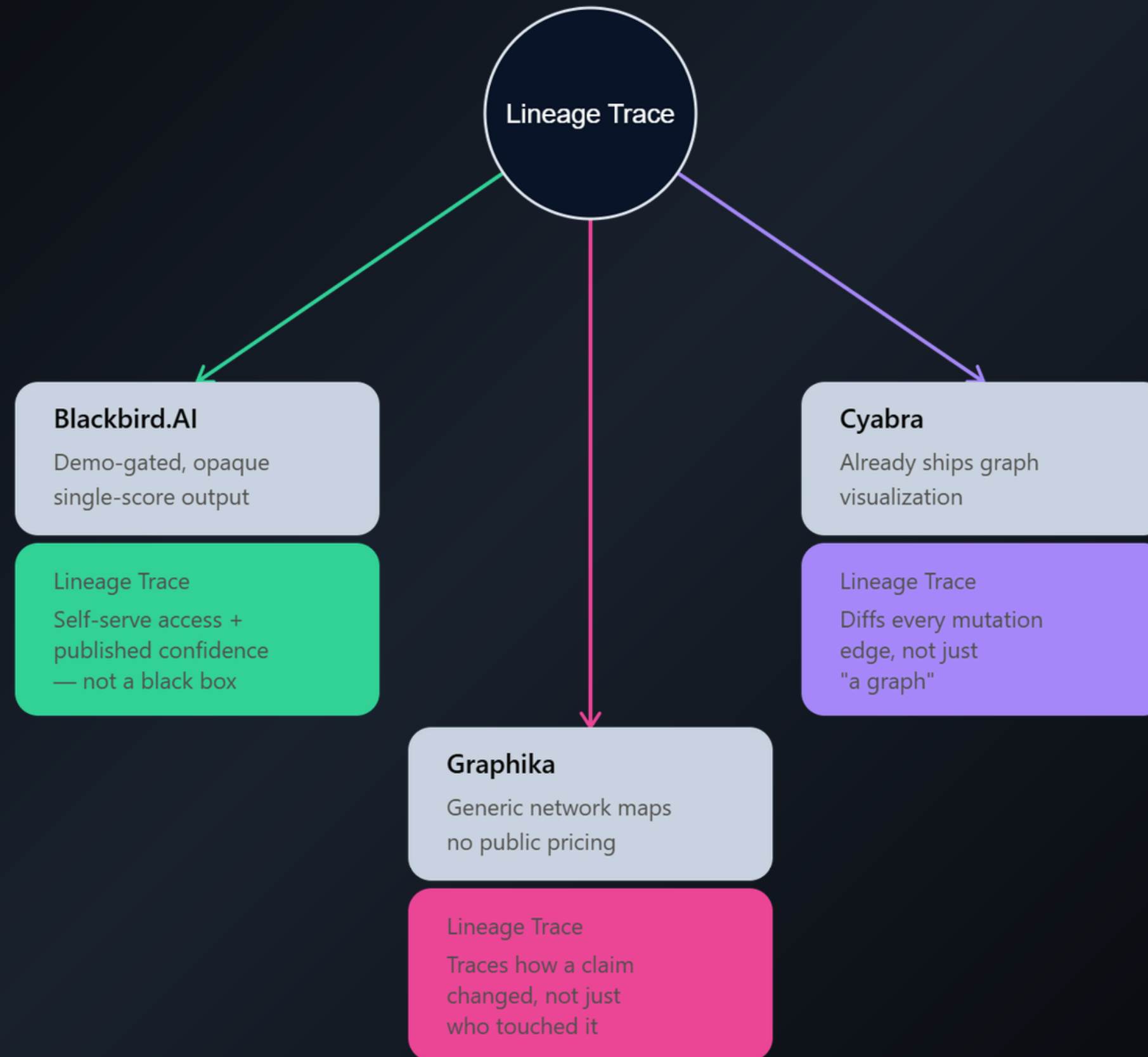
PROPOSED SOLUTION

- Traces lineage, not just similarity: clusters claims by meaning using multilingual embeddings, so paraphrased or reworded versions stay one traceable chain, not scattered as unrelated posts
- Diffs every mutation edge: token-level alignment flags exactly which words, numbers, or context shifted at each hop in the spread
- Scores mutation danger: each edit is LLM-rated for severity — factual distortion, inflammatory language added, lost context — producing a danger_delta, not just a "different" flag
- Classifies spread topology: combines graph shape with timing signals (NetworkX) to label a cluster organic, coordinated, or bot-amplified
- Computes R_claim: a velocity metric — new mutation edges ÷ active nodes, tracked in windowed intervals — that flags acceleration early
- Tracks debunk-lag: measures edges and time between a correction entering the graph and the mutation rate actually dropping, not just whether one was posted

TARGET USERS

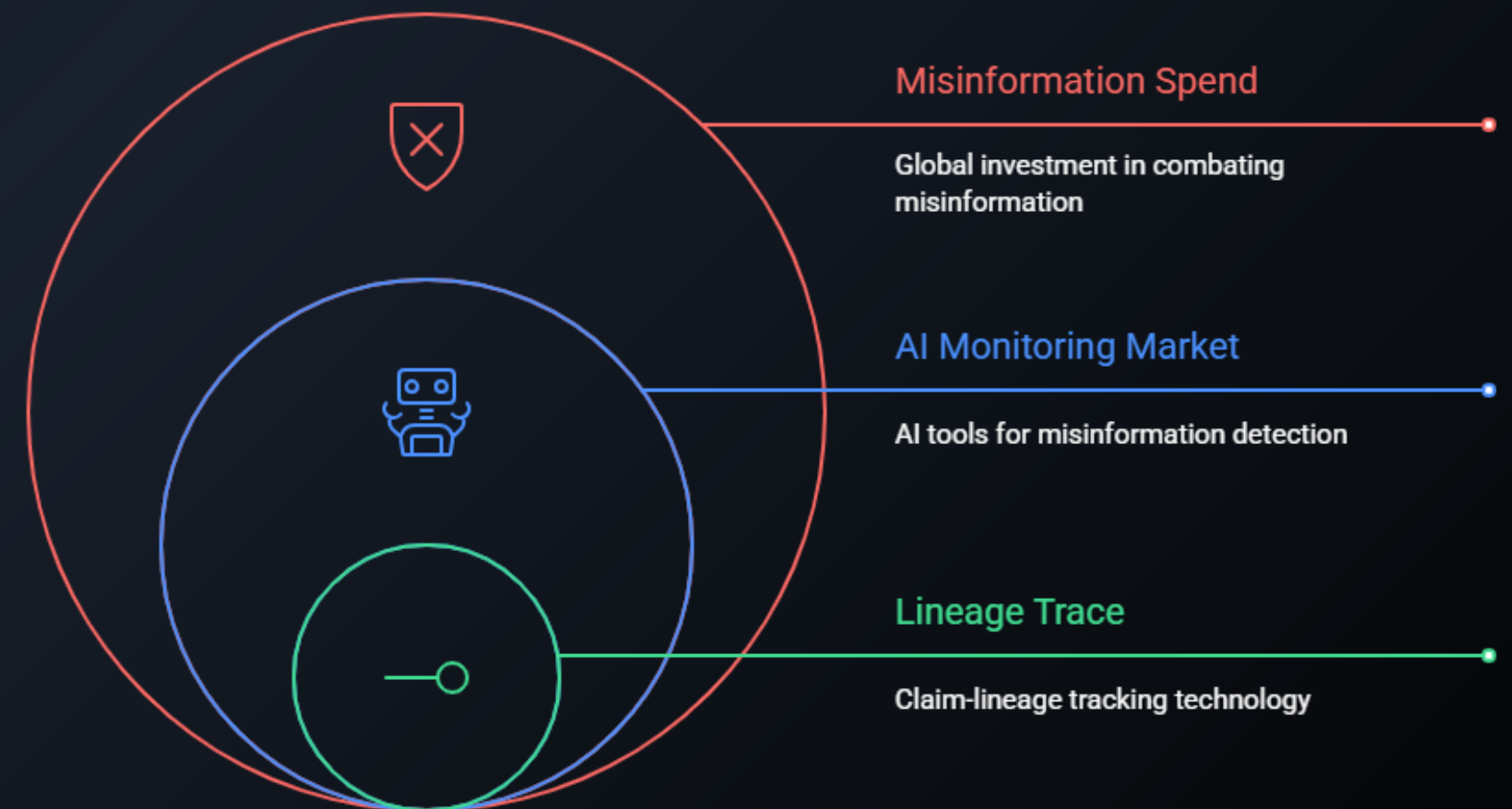
Segment	Who They Are	Current Pain Point	Expectations from Lineage
Newsroom Standards Desks	Editors/fact-checkers verifying claims pre-publication	Manual tracing takes too long — story moves on before the trace is done	Fast, visual origin + mutation reconstruction, citable in a correction
Political Monitoring Teams	Campaign/war-room staff tracking narrative attacks	Reactive - alerted only after a narrative has traction, no view of which version is driving it	Early-warning velocity (R_claim) + the specific "dangerous" mutation
Brand Reputation Teams	Corporate comms/PR managing attacks on a company or exec	Legacy tools report 6–48hr detection lags	Near-real-time detection + a map of who amplified what

UNIQUE VALUE PROPOSITION



MARKET OPPORTUNITY

- Enterprise misinformation spend projected to surpass \$30B by 2028 (Gartner); WEF ranks it a top 2026 global risk. Adjacent AI monitoring market: ~\$6.5B (2025) → \$23B (2034).
- Funded players exist (Blackbird.AI, Cyabra, Graphika, NewsWhip, Logically), but legacy tools run 6–48hr detection lags and none lead with claim-level mutation tracking. That's Lineage Trace's wedge.



PRODUCT CONCEPT

Ingestion & understanding

Monitor channels via Telegram Bot API
Cluster claims via Sentence-Transformer embeddings

Lineage & reasoning

Build mutation graphs with NetworkX
Diff edges, score danger, classify topology
(organic / coordinated / bot) via Anthropic API

Metrics & serving

Calculate R_claim and debunk-lag
Serve via FastAPI, visualize with D3.js

Node: claim_id · language · timestamp · platform | Edge: mutation_type · diff_score · danger_delta

TECH STACK



TELEGRAM API

Live claim ingestion via direct access to the platform where regional-language misinformation originates



SENTENCE TRANSFORMER

Multilingual embeddings for clustering that keep a claim's lineage intact across languages



NETWORKX

Mutation tree and topology metrics using graph math to scale to large cascades.



ANTHROPIC API

Mutation diffing and danger-score reasoning reserved for steps requiring analysis.



FAST API

Pipeline orchestration that is light-weight and fast to wire up for a real API layer



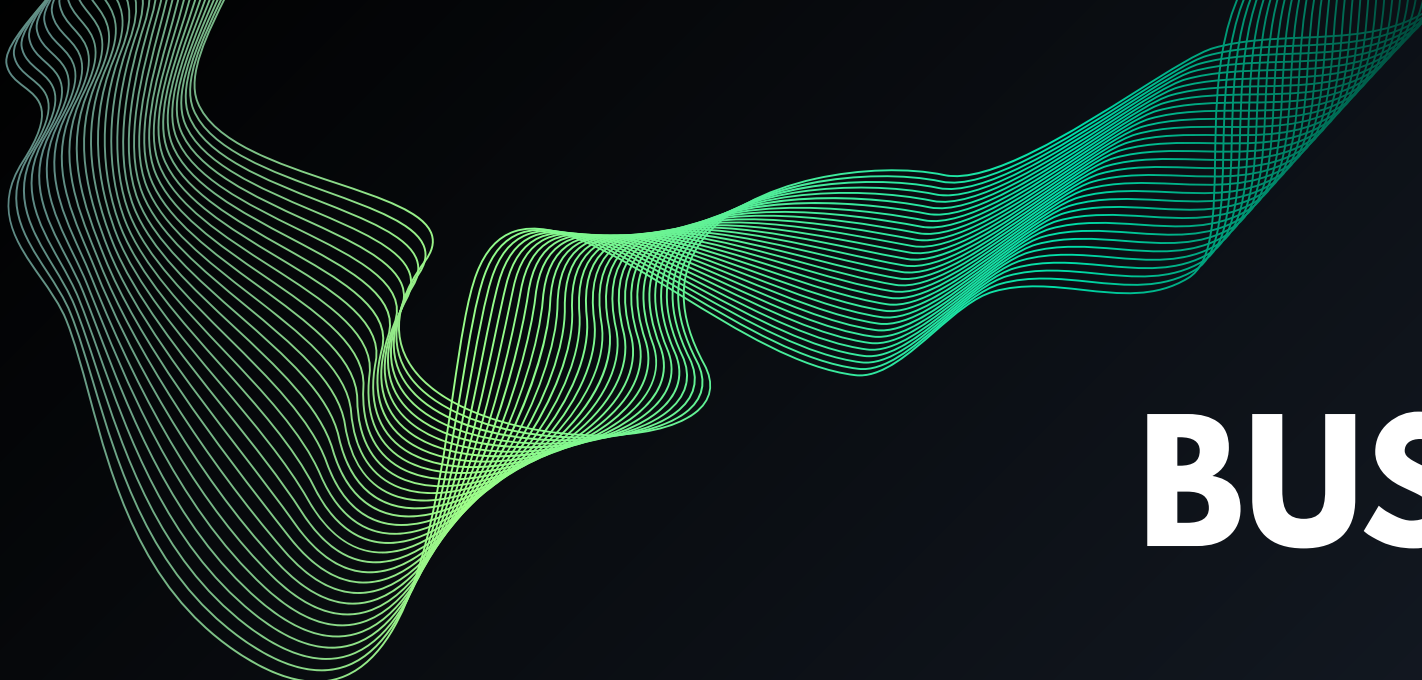
D3.JS GRAPH

Mutation river rendering for the interactive, force-directed visual



REPLIT

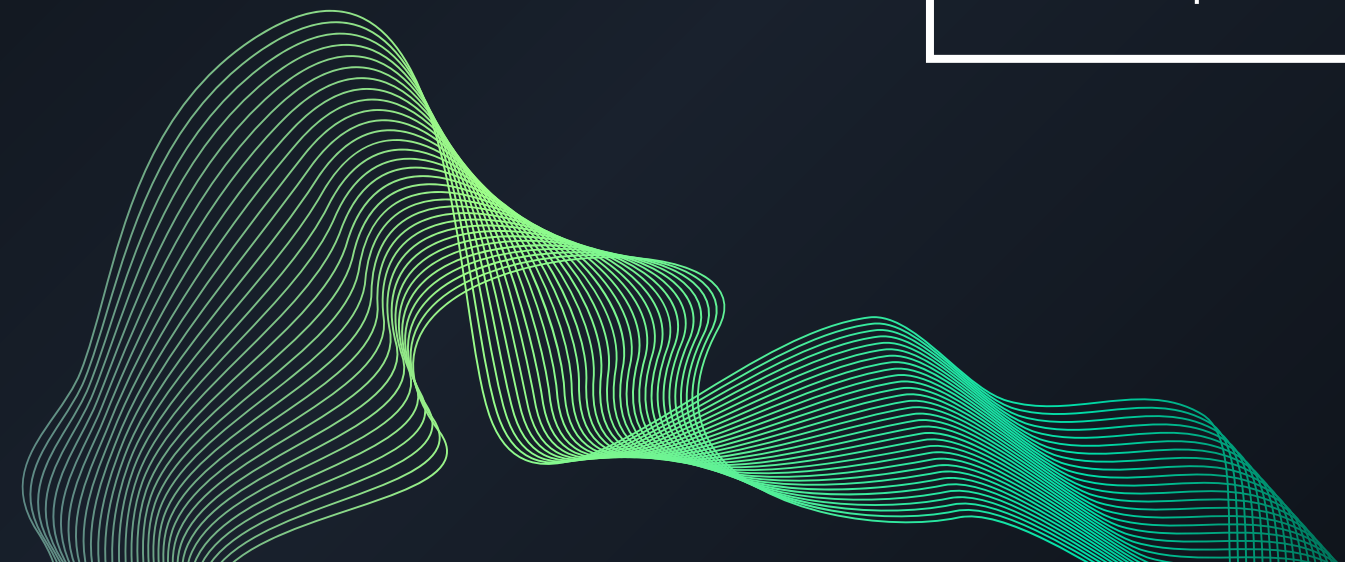
Hosting platform used to de-ploy and demo live without infrastructure setup time.



BUSINESS MODEL

- Institutional buyers → seat-based B2B SaaS, priced against monitoring capacity (topics tracked, seats, alert volume), not per-query.
- Value compounds with use — richer mutation history over time is a natural expansion path from Standards → Enterprise.

Tier	Price	Includes
Newsroom Standards	~\$500–1,500/mo	Single-team access, on-demand tracing, alert digests
Political Monitoring	~\$2,000–5,000/mo	Multi-seat, real-time ingestion, custom watchlists
Brand & Enterprise	Custom pricing	API access, white-label reports, SLA-backed alerting



FUTURE ROADMAP

Vision: Advanced Tracing and Expansion

Lightweight verdict layer with lineage tracing; node typing; expansion beyond Telegram to X and WhatsApp.

Now: Core Functionality

Mutation river and topology classifier live on seeded data; danger scoring, R_claim, and core diffing operational.

Next: Integration and Clarification

Live Telegram ingestion wired; edge-level coordination attribution; danger score and R_claim units clarified.

